

Project Design Phase Solution Architecture

Date	28-06-2026
Team ID	24A55A4203
Project Name	Visitor management system
Maximum Marks	4 Marks

Solution Architecture:

The Visitor Management System is engineered on the ServiceNow enterprise platform utilizing a highly structured, decoupled, and secure three-tier application architecture. This setup ensures seamless transaction isolation, precise security control via system roles, and immediate data persistence parsing across custom scoped tables:

User Interaction Layer (Service Portal / Kiosk UI): Provides an intuitive self-service portal layout or front-desk lobby kiosk context. Employee hosts configure visitor names, target email profiles, phone records, and scheduled meeting dates via native Catalog Items.

Business Logic Layer (Client & Server Engines): Coordinates incoming operations via Catalog Client Scripts, Client-callable Script Includes, and asynchronous GlideAjax calls. This engine handles input evaluation rules (e.g., telephone string validation patterns) and executes interactive state updates via UI Actions.

Data & Integration Service Layer (Cloud Schema & Assets): Manages record persistence across custom application tables (such as x_snc_visitor_log). This layer controls security filters via Access Control Lists (ACLs), triggers transactional email alerts, embeds virtual QR code validation assets, and tracks live security metrics using graphical workspace dashboards.

Solution Architecture Diagram:

